

DUAL-CORE OPTICAL MOTION TESTBED

MECHANICAL PLATFORM & INTERFACE PCB

Rotary/translation mechanics, differential encoder interface, STEP/DIR control, grounding, and physical verification

ENGINEERING REQUIREMENTS DOCUMENT | REVISION 5.1 | 5 AUGUST 2026

Prepared by	Herder Elektronische Systemen
Document scope	Mechanical platform, drivetrain, custom interface PCB, grounding, assembly, and bench verification
Evidence basis	CAD models, fabrication photographs, schematics, PCB routing, ground-plane views, load demonstrations, and test records
Distribution	Customer-facing; controlled source excerpts and engineering evidence

DOCUMENT PURPOSE
Defines the controlled functional, performance, interface, safety, verification, and acceptance requirements for the mechanical platform and encoder/STEP-DIR interface PCB. Firmware behavior and quantitative control results are referenced to the companion firmware document.

Requirements Basis

Requirements Definition and Verification

Requirements are uniquely identified and paired with rationale, interfaces, constraints, verification methods, and acceptance evidence. Demonstrated prototype performance is distinguished from unverified, unrated, or future production capability.

Purpose, Scope, and System Boundary

This FRD governs the mechanical platform, rotary/translation architecture, physical interfaces, encoder/STEP-DIR PCB, grounding strategy, assembly evidence, and bench-level verification. Target firmware, GUI behavior, and quantitative control results are external interfaces documented in the companion firmware FRD.

Boundary element	Included in this FRD	Excluded / referenced
Mechanical platform	Frame, rotary stage, translation mechanism, top plate, drivetrain, assembly	Target control algorithms and GUI implementation
Interface PCB	Differential receiver, connectors, routing, dedicated ground plane, assembly views	Production certification and released manufacturing package
Verification	Travel observation, alignment references, static load demonstration, integration videos	Rated payload, environmental qualification, EMC, and optical metrology certification

Users and Operational Scenarios

Actor	Primary interaction	Expected outcome
Operator	Mount hardware, connect power/control, establish position reference, run motion	Accessible and stable bench-scale test operation
Mechanical engineer	Inspect structure, travel, stiffness, assembly, and serviceability	Traceable platform architecture and known design boundaries
Electrical engineer	Inspect schematic, routing, ground plane, and connector functions	Reviewable signal-conditioning and grounding intent
Reviewer/customer	Compare requirements with photographs, CAD, PCB views, and tests	Clear evidence of what is built, demonstrated, and not certified

Functional Requirements

ID	The platform shall...	Rationale / explanation	Verification
PL-FR-001	provide integrated rotary and translational motion on a rigid bench-scale frame.	The test article must support combined optical-motion experiments in one serviceable assembly.	CAD review; physical inspection; demonstration video
PL-FR-002	provide more than plus/minus 3 inches of demonstrated translational travel.	The translation axis must cover the intended bench adjustment range.	Ruler/needle observation and travel photographs
PL-FR-003	support the demonstrated 40 lb static offset load without loss of assembly integrity.	The platform must carry representative mounted equipment during bench evaluation.	Static load demonstration; visual inspection

ID	The platform shall...	Rationale / explanation	Verification
PL-FR-004	provide differential encoder reception and logic-level delivery to the controller.	Differential signaling reduces susceptibility to cable-borne common-mode noise.	Schematic review; AM26LS32 identification; integration test
PL-FR-005	route STEP, DIR, encoder, power, and ground through identifiable connectors.	Field wiring must remain inspectable, serviceable, and resistant to ambiguous connections.	PCB/schematic inspection and continuity check
PL-FR-006	use a dedicated solid-copper ground plane beneath the routed signal layer.	A continuous return reference reduces fragmented current paths and supports signal integrity.	PCB layer review and fabrication-file inspection
PL-FR-007	permit inspection and service of mechanical and electrical subassemblies.	Prototype maintenance and revision require visible fasteners, connectors, and modular components.	Assembly review and service-access inspection
PL-FR-008	provide a physical emergency-stop control accessible to the operator.	The operator needs an immediate hardware-level intervention point.	Physical inspection and stop-function test

Performance and Quality Requirements

ID	Requirement	Acceptance basis
PL-NFR-001	The frame and top plate shall remain visibly intact and assembled during the documented 40 lb static test.	No visible structural separation or loss of support during the demonstration.
PL-NFR-002	The translation mechanism shall demonstrate travel beyond plus/minus 3 inches.	Photographic/ruler evidence documents the achieved range.
PL-NFR-003	Differential encoder inputs shall be conditioned through the identified receiver stage before reaching controller logic.	Schematic and populated-board evidence agree.
PL-NFR-004	The routed layer and dedicated ground plane shall be shown as separate, reviewable PCB layers.	Routing and plane figures are both present and consistent with the schematic.
PL-NFR-005	Prototype wiring and connectors shall remain accessible for inspection and troubleshooting.	Physical and PCB images show identifiable, reachable connection points.
PL-NFR-006	The report shall not represent the static load, motion evidence, or PCB as a certified product rating.	Limitations clearly distinguish demonstration from certification.

External Interface Requirements

Interface	Required behavior	Failure / boundary condition
Motor driver	Accept STEP/DIR commands and drive the geared stepper mechanism.	Incorrect polarity or command wiring shall be prevented through labeling and controlled harnessing.
Encoder	Deliver differential position channels to the receiver PCB.	Open, swapped, or noisy channels shall be diagnosable at the PCB/controller interface.
STM32H747 controller	Receive conditioned encoder logic and	Controller and PCB grounds must share the

Interface	Required behavior	Failure / boundary condition
	provide control outputs.	defined reference.
Power input	Supply the motor/control hardware through protected, identifiable connections.	Final current, peak power, and production protection remain release actions.
Mounted test article	Attach to the top plate within the demonstrated load and geometry envelope.	The 40 lb demonstration is not a certified payload rating.

1. Platform Overview

The platform combines a rigid aluminum frame, rotary stage, translation mechanism, optical-table-style upper surface, geared stepper drive, and differential encoder feedback. The custom interface PCB conditions encoder signals and organizes STEP/DIR, control, and power connections between the motion hardware and the STM32H747 target.

2. Mechanical Architecture

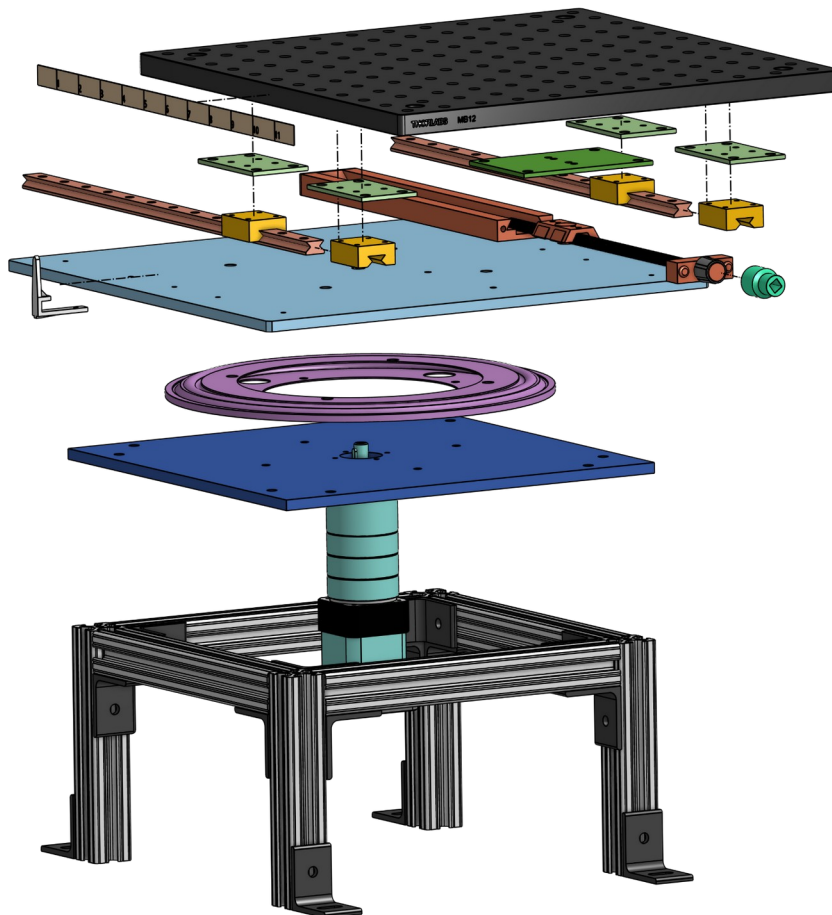


Figure 1. Exploded CAD view showing frame, rotary stage, translation rails, upper plate, and drive components.

- Rotary and translational degrees of freedom are integrated into one bench-scale assembly.
- The frame uses modular aluminum extrusion and a lower base plate for stiffness and serviceability.
- The top plate supports optical fixtures and controlled offset loading.
- The translation mechanism provides more than ± 3 inches of travel in the demonstrated configuration.

3. Fabrication and Assembly Evidence



Figure 2. In-progress assembly showing the frame and mechanical integration.

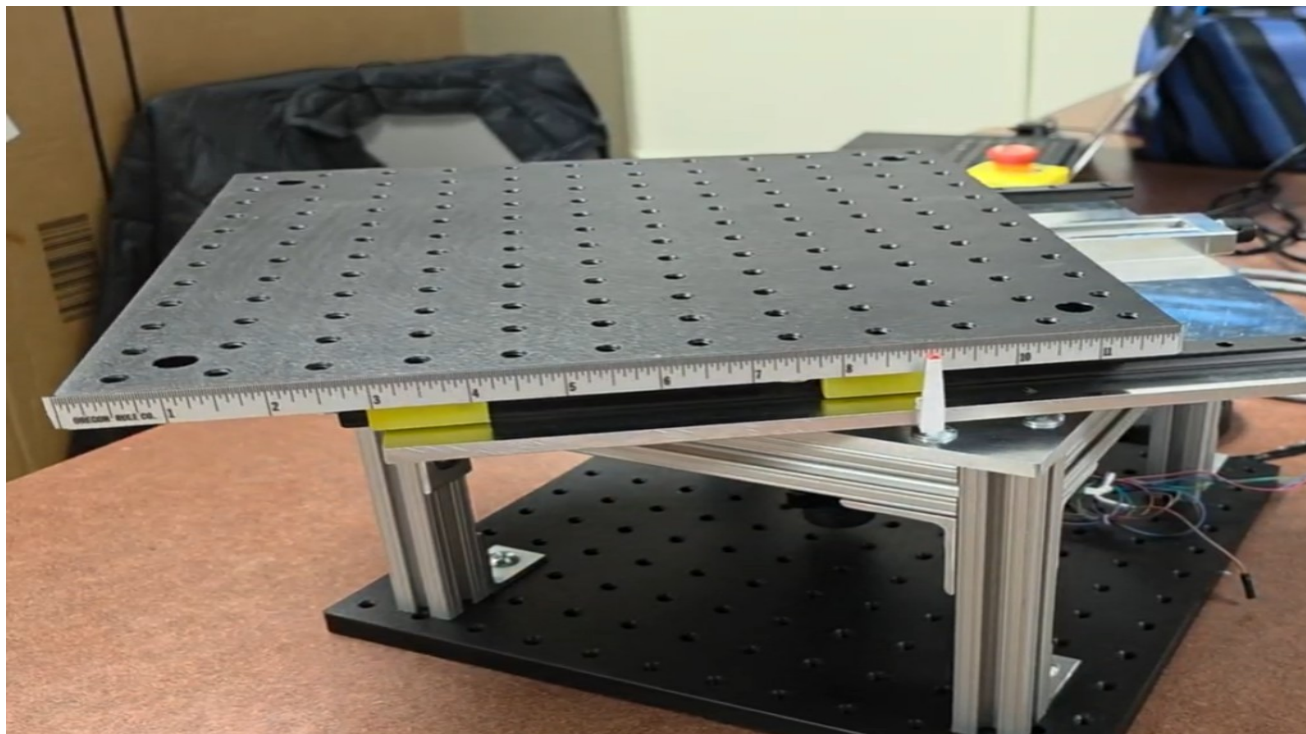


Figure 3. Final assembled testbed configuration.

4. Translation and Alignment Verification

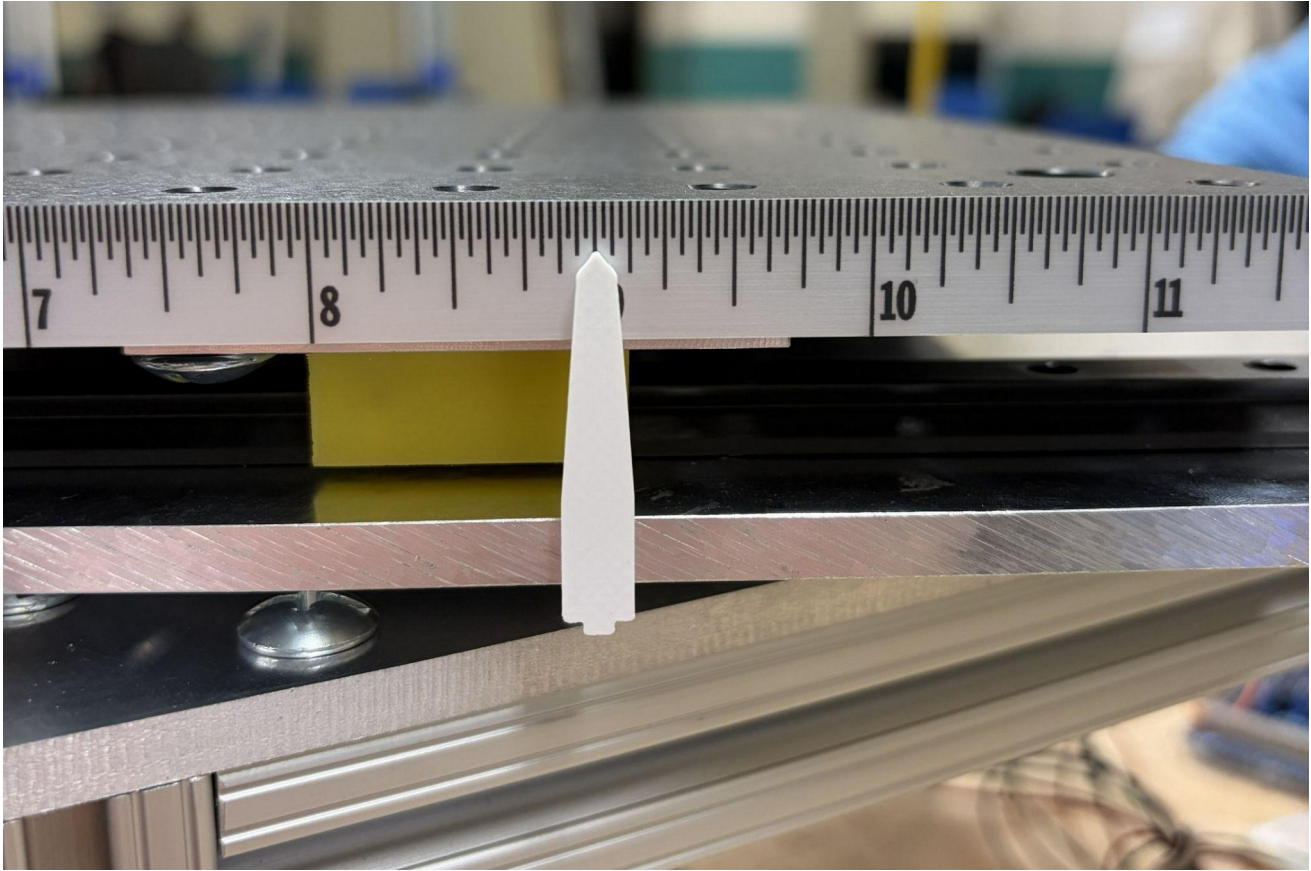


Figure 4. Needle-based translation reference used to observe displacement and alignment.

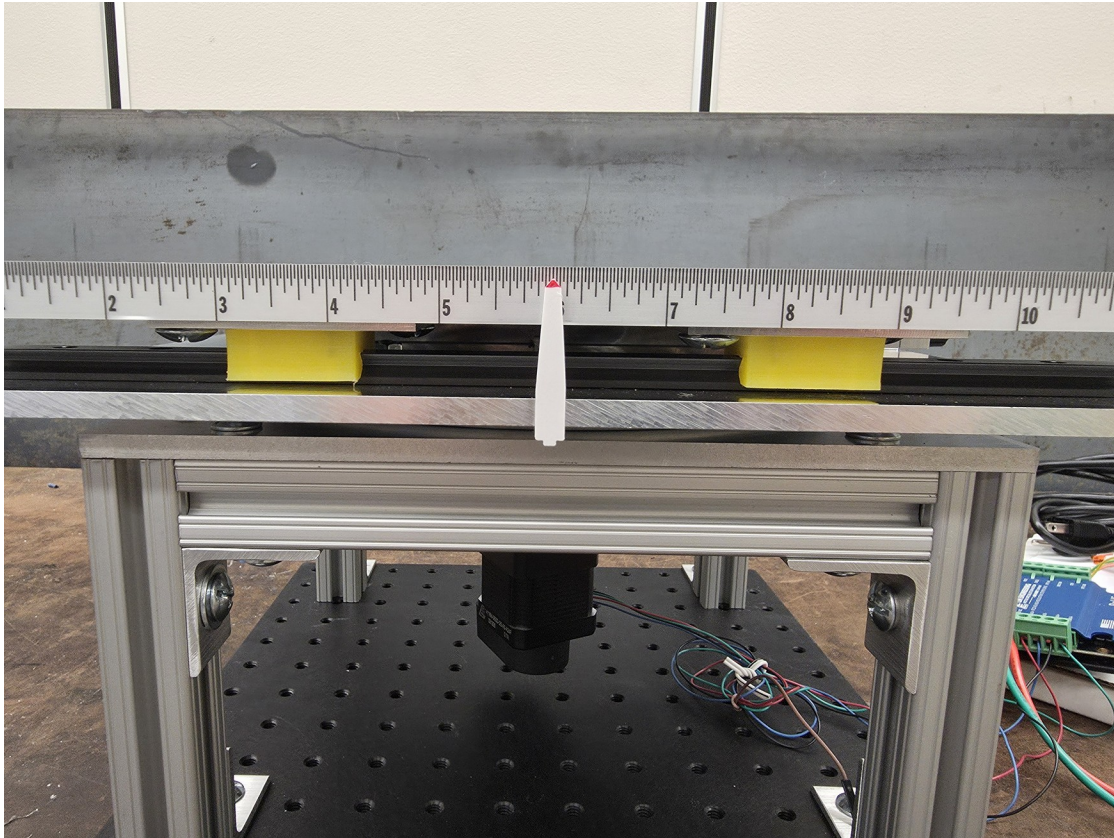


Figure 5. Centered translational precision needle during physical verification.

5. Static Load Demonstration



Figure 6. Forty-pound static offset-load demonstration on the completed platform.

Condition	Observed evidence	Scope
Centered load	Platform supports the applied static mass	Demonstration, not certification
Offset load	Motion hardware remains assembled and functional	Representative bench test
Translation reference	Needle and ruler provide visible displacement reference	Qualitative/bench verification

6. Custom Encoder / STEP-DIR Interface PCB

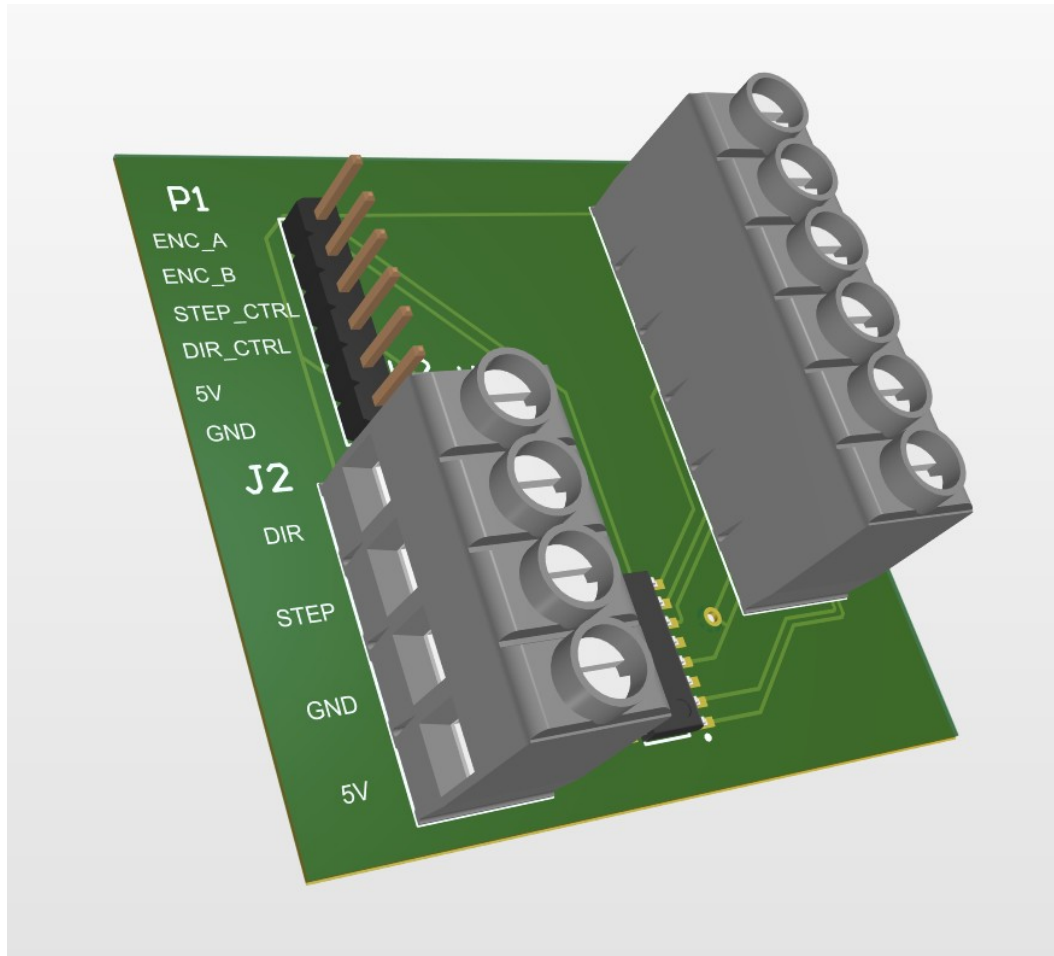


Figure 7. Populated 3D rendering of the custom interface PCB.

Circuit element	Function
AM26LS32 differential receiver	Converts differential encoder channels to logic-level signals
Encoder connector	Routes A/B and related encoder signals
STEP/DIR connector	Routes motion commands and control references
Controller header	Interfaces the board with the STM32H747 target
Terminal blocks	Provide field-wiring access for encoder, control, power, and ground

7. Schematic and Signal Routing

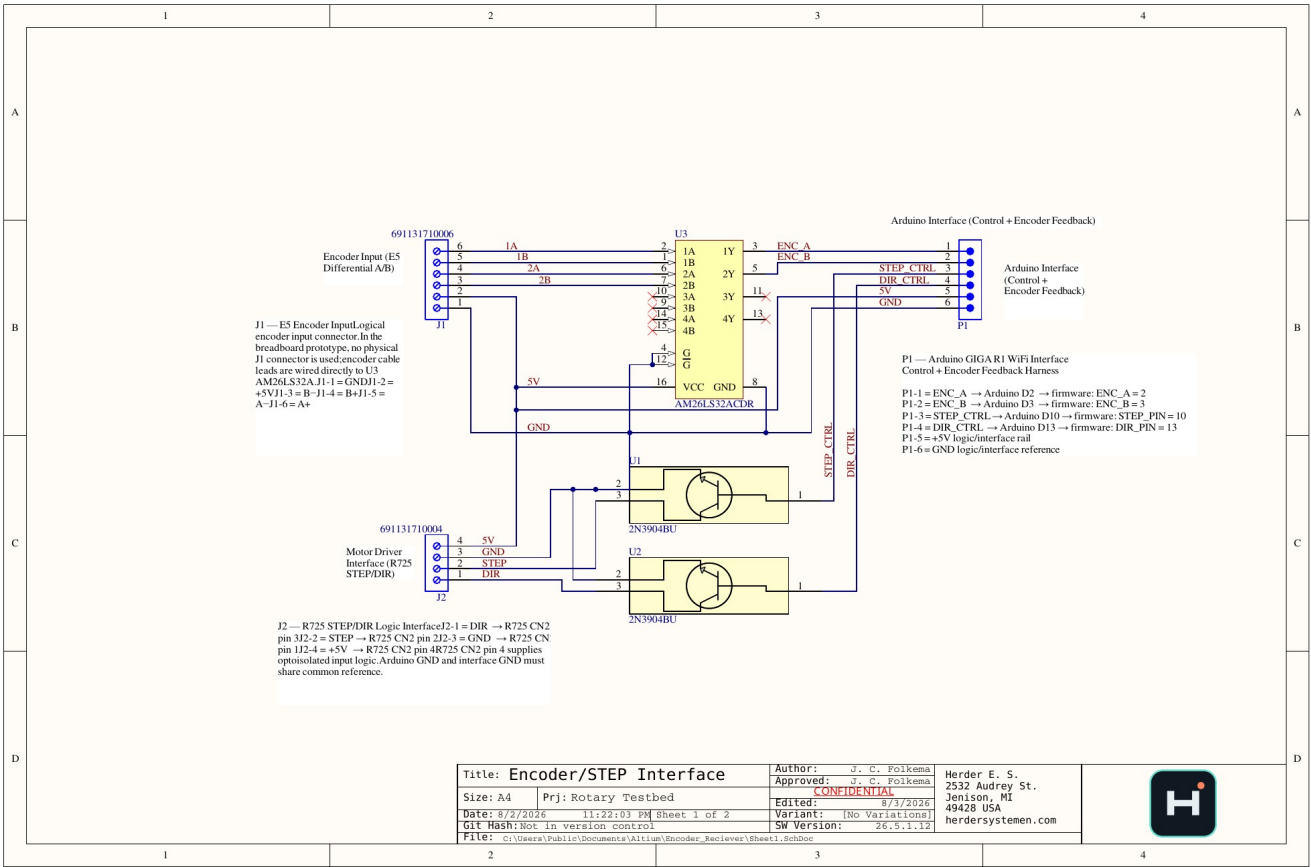


Figure 8. Encoder and STEP/DIR interface schematic.

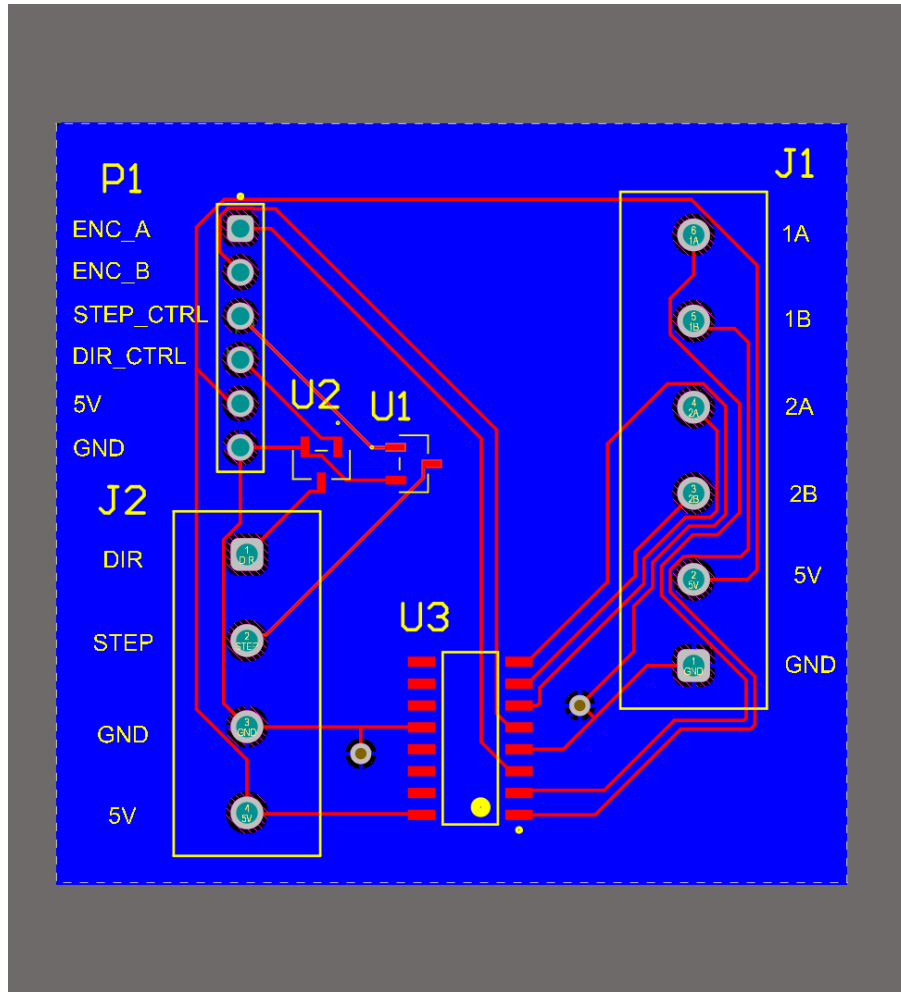


Figure 9. Routed signal layer with component placement and connector organization.

8. Dedicated Ground Plane

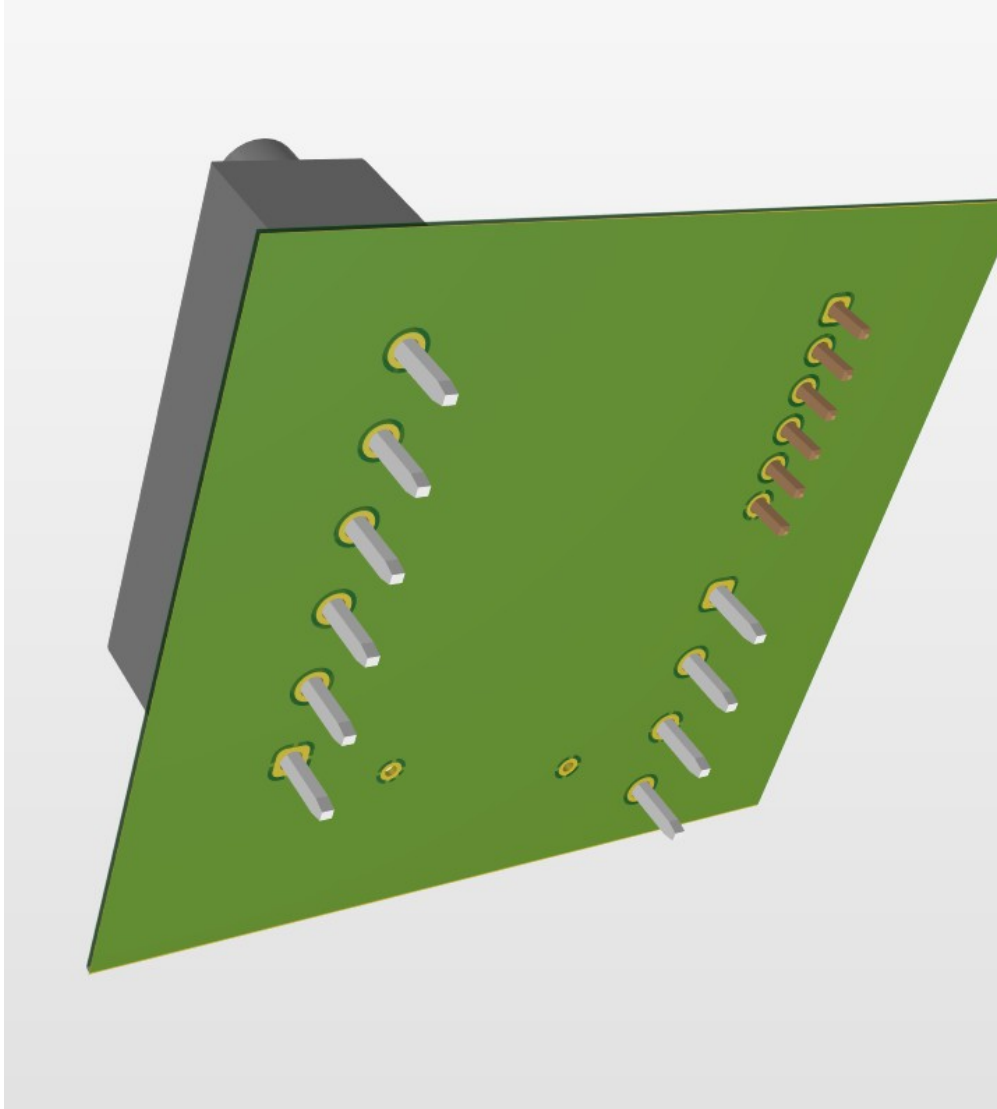


Figure 10. Dedicated solid-copper ground plane on the opposite PCB layer.

The dedicated ground plane provides a continuous return reference beneath the routed signal layer. This is particularly important for differential encoder reception, controller reference stability, and minimizing fragmented return paths across the compact interface board.

9. PCB Assembly Views

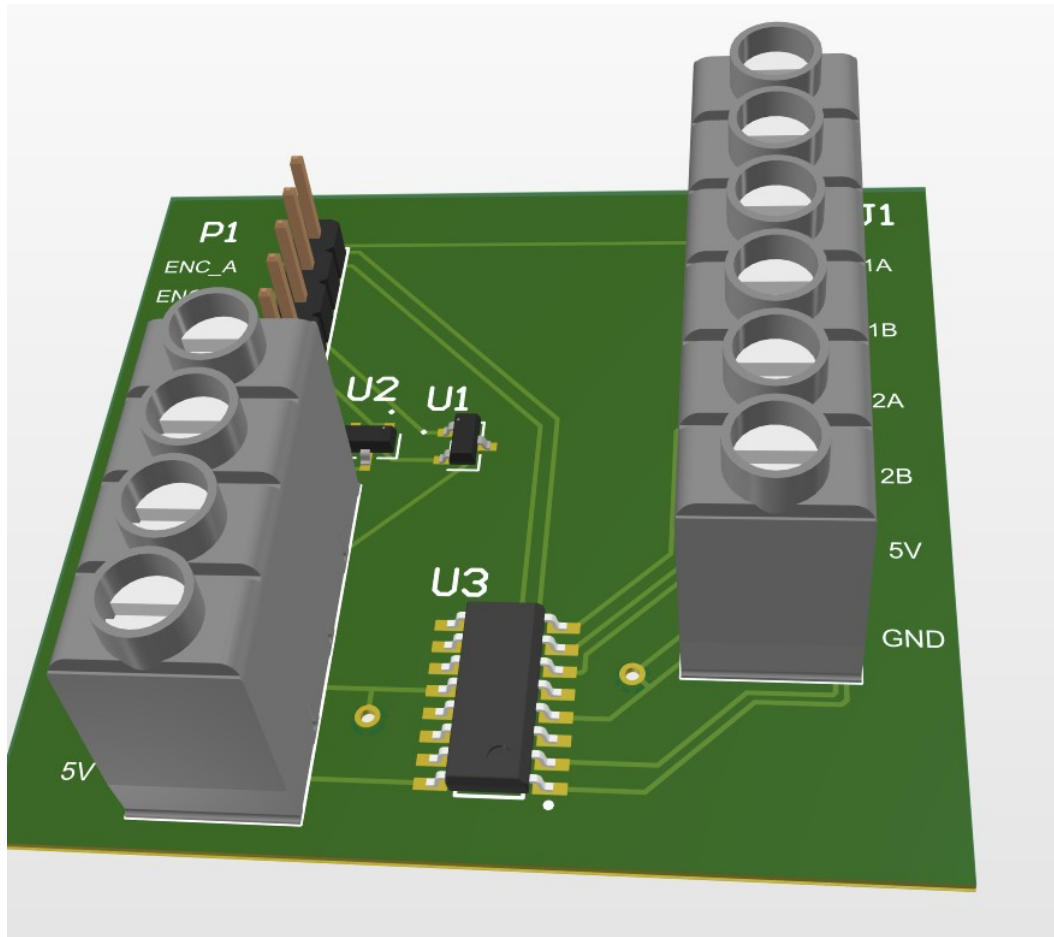


Figure 11. Angled PCB assembly rendering showing terminal blocks, receiver IC, and headers.

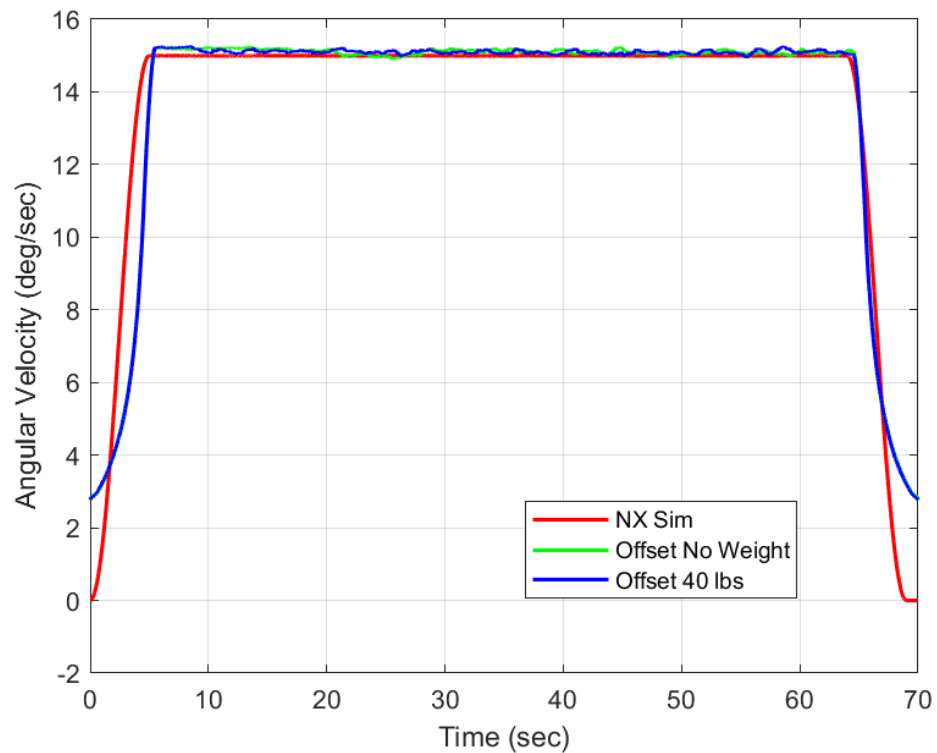


Figure 12. Alternate PCB assembly view.

10. Dynamic Verification Evidence

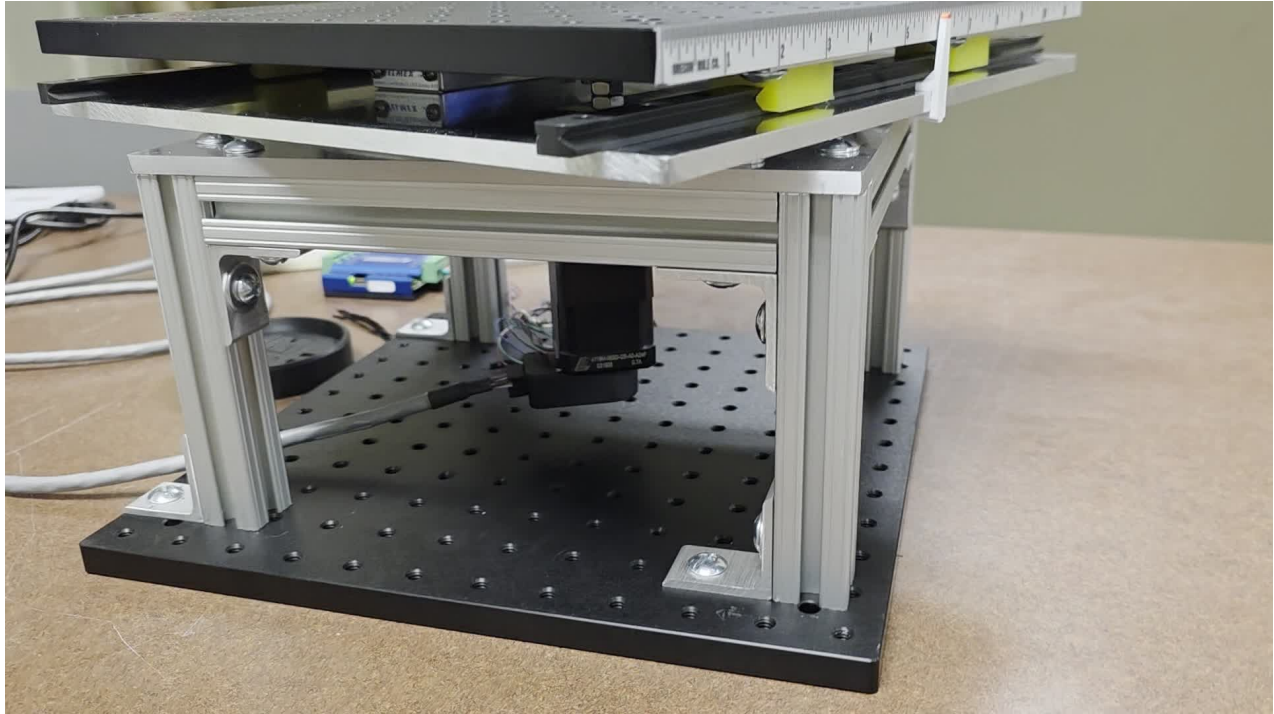
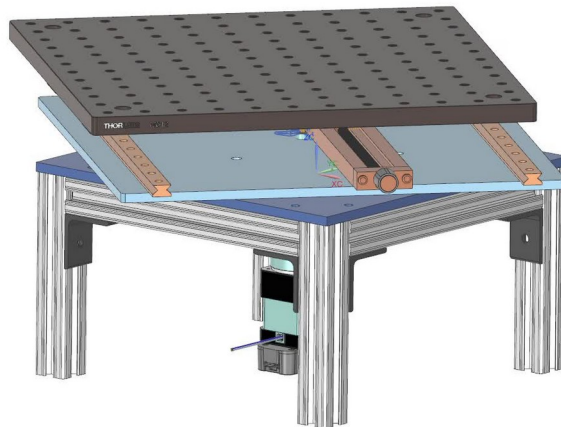


Figure 13. Frame from the integrated project-solution demonstration video.

Solution1	
Time	5.880000
Step	588



Solution1 Default Animation

Figure 14. Frame from the NX model animation used for mechanism review.

Recorded demonstrations include the integrated project solution video, an NX model animation, centered and offset load videos, and motion demonstrations. The companion firmware report contains the quantitative control and telemetry results; this report limits its conclusions to physical architecture and observed bench verification.

11. Detailed Mechanical and Electrical Design

The image set is supported by the following design detail so the report remains an engineering record rather than a photograph gallery. The values below consolidate the demonstrated mechanical envelope, drivetrain, encoder interface, PCB architecture, and verification evidence.

Area	Documented implementation	Design intent / evidence
Mechanical frame	Modular aluminum extrusion structure with lower base plate and optical-table-style upper plate	Rigid, serviceable bench platform supporting rotary and translational motion.
Translation	Demonstrated travel greater than +/-3 inches	Needle references and centered-position photographs document alignment and displacement.
Static loading	40 lb offset load demonstrated on the completed assembly	Confirms the prototype can support representative off-axis payloads without visible structural failure.
Rotary drive	200-step motor, 256 microsteps, 24.65:1 gearing	1,262,080 command pulses per table revolution.
Encoder feedback	5,000 CPR differential encoder; 10,000 decoded counts/rev at the motor	246,500 table counts/rev after gearing.
Controller interface	STM32H747-based controller using STEP/DIR output and differential encoder input	Connects the mechanical platform to the dual-core motion and telemetry firmware.

11.1 Custom Encoder / STEP-DIR PCB

Circuit element	Implementation	Function
Differential receiver	AM26LS32-class line receiver	Converts differential encoder channels to logic-level signals with improved noise immunity.
Encoder connector	Differential A/B/index and reference connections	Provides organized field wiring between the encoder and receiver circuitry.
Controller headers	Logic-level encoder outputs and STEP/DIR/control signals	Interfaces the custom board with the STM32H747 controller.
Driver connections	STEP, DIR, enable/reference, and supply terminals	Routes command signals to the external stepper drive.
Signal layer	Purpose-routed traces with separated connector zones	Keeps encoder reception, controller I/O, and drive command routing legible and serviceable.
Ground layer	Dedicated continuous solid-copper plane	Provides a low-impedance return reference beneath the signal layer and reduces fragmented return paths.

11.2 Ground-Plane and Signal-Integrity Rationale

- The solid plane is a separate PCB layer, not a cosmetic copper pour on the routed layer. It maintains a continuous return path beneath encoder and control traces.
- Differential signaling improves rejection of common-mode noise on the encoder cable, while the receiver establishes clean logic-level edges for the controller.
- Connector grouping and a visible one-layer routing strategy make the prototype easier to inspect, troubleshoot, and revise without obscuring the grounding approach.
- The report intentionally shows the schematic, routed signal layer, dedicated ground plane, and multiple 3D assembly views so electrical intent can be checked against physical implementation.

11.3 Verification Evidence and Configuration Boundaries

Verification item	Evidence retained	Boundary
Assembly	Exploded CAD, fabrication photographs, and final cropped assembly image	Prototype build evidence; not a production manufacturing release.
Translation	Needle-reference photographs at centered and displaced positions	Demonstrates travel and alignment qualitatively; not a calibrated metrology certification.
Load	40 lb static offset-load photograph	Static demonstration only; no formal shock, vibration, fatigue, or long-duration qualification.
Electrical	Schematic, routing, ground plane, and populated PCB renders	Design evidence; editable manufacturing files remain controlled.
Integrated motion	Project-solution video frame and NX animation frame	Demonstrates integration and mechanism review; quantitative motion results are reported in the firmware FRD.

Requirements Verification and Traceability

The matrix below links each requirement to objective evidence and a current disposition. “Demonstrated” means evidence is present in this report or its cited source package; “Release action” means the requirement remains to be completed or formally witnessed before a controlled production release.

Requirement	Evidence / verification artifact	Disposition
PL-FR-001	Exploded CAD and completed assembly images	Demonstrated
PL-FR-002	Translation reference photographs	Demonstrated
PL-FR-003	40 lb offset-load photograph/video	Demonstrated, not certified
PL-FR-004	Schematic, populated PCB, receiver identification	Demonstrated
PL-FR-005	Schematic, routing, connector views	Demonstrated
PL-FR-006	Dedicated ground-plane figure	Demonstrated
PL-FR-007	Fabrication and assembly photographs	Demonstrated
PL-FR-008	Physical emergency-stop visible on platform	Demonstrated; functional witness recommended
PL-NFR-001 to 006	Mechanical/PCB evidence and limitations	Mixed: demonstrated and controlled limitations

Acceptance and Release Criteria

- The platform remains mechanically assembled through the defined travel and static-load demonstrations.
- Encoder, STEP/DIR, power, and ground connections pass documented continuity and polarity checks.
- Schematic, routed layer, ground plane, and populated-board configuration agree with the controlled fabrication revision.
- The emergency-stop and motion interfaces are functionally witnessed before customer deployment.
- Payload, environmental, EMC, and optical-performance claims remain explicitly outside the demonstrated prototype scope.

12. Limitations and Document Control

Item	Controlled statement
Prototype manufacture	No formal shock, vibration, thermal, EMC, or long-duration certification is claimed.
Electrical power	Idle, steady-state, and peak system power remain to be measured.
Fabrication release	Editable board files and the final manufacturing package remain controlled.
Metrology	Motion-platform characterization does not certify downstream optical performance.
Load evidence	The 40 lb result is a static demonstration, not a rated payload certification.

Attribution

Herder Elektronische Systemen developed the public-facing engineering package and documents the platform architecture, custom encoder/STEP-DIR PCB, physical verification evidence, and release controls represented here. No third-party affiliation or endorsement is claimed.

Revision record

Revision	Date	Description
5.0	2026-08-05	Reframed as a controlled FRD with explicit functional requirements, interfaces, acceptance criteria, traceability, while preserving mechanical, schematic, routed-copper, ground-plane, PCB-render, and physical-verification imagery.

Configuration and Reproduction

This section defines the switch settings, physical connector map, prototype electrical implementation, setup sequence, and acceptance checks needed to reproduce the demonstrated hardware configuration.

R725 Microstep DIP-Switch Table

uSTEP	SW1	SW2	SW3	SW4	Pulses/motor rev	Recorded output speed
2	U	U	U	U	400	0.02608 deg/s
4	D	U	U	U	800	0.01304 deg/s
8	U	D	U	U	1,600	0.00652 deg/s
16	D	D	U	U	3,200	0.00326 deg/s
32	U	U	D	U	6,400	0.00163 deg/s
64	D	U	D	U	12,800	0.000815 deg/s
128	U	D	D	U	25,600	0.000407 deg/s
256	D	D	D	U	51,200	0.000204 deg/s
5	U	U	U	D	1,000	0.01043 deg/s
10	D	U	U	D	2,000	0.00522 deg/s
20	U	D	U	D	4,000	0.00261 deg/s
50	D	D	U	D	10,000	0.00104 deg/s
100	U	U	D	D	20,000	0.000522 deg/s
200	D	U	D	D	40,000	0.000261 deg/s
18	U	D	D	D	3,600	0.00290 deg/s
180	D	D	D	D	36,000	0.000290 deg/s

U and D reproduce the notation in the source workbook. The characterized low-speed configuration used 256 uSTEP: SW1 D, SW2 D, SW3 D, SW4 U. Switch state must be verified against the physical R725 labeling before energizing the axis.

Motor-Current and Standby-Current Configuration

The current-selection switches and standby-current option shall be recorded for every released bench configuration. The source package includes the R725 manual, but it does not unambiguously identify which current switch combination was physically used during every recorded test. The release record shall therefore contain: selected peak/phase current, current-switch positions, standby-current setting, motor rated current, and operator/date. No unsupported current setting is asserted in this FRD.

Seven-Pin Interface Connector

Pin	Signal	Controller mapping	Status	Physical route / note
1	Encoder A	D2 / PA3	Active	AM26LS32 receiver output to controller
2	Encoder B	D3 / PA2	Active	AM26LS32 receiver output to controller
3	STEP	D10 / PK1	Active	Controller through 2N3904 interface to R725 STEP
4	DIR	D13 / PH6	Active	Controller through 2N3904 interface to R725 DIR
5	Encoder Z / Index	D4 / PJ8	Wired; unused	Receiver output present; current firmware does not consume it
6	+5 V	5 V	Power	Logic/interface rail
7	Ground	GND	Power	Common interface reference; documented lower main-header ground

Prototype Electrical BOM

Designator / location	Function	Part	Qty	Implementation status
U3	Differential encoder receiver	TI AM26LS32ACN, PDIP-16	1	Used in prototype
U1, U2	STEP and DIR interface stages	onsemi 2N3904BU, TO-92	2	Used in prototype
Breadboard wiring	Direct interconnects	Male-male / solid-core jumpers	1 set	Used in prototype
Breadboard	Temporary assembly platform	Solderless breadboard	1	Used in prototype
WAGO splices	Harness distribution	WAGO 221-2401 or equivalent	As required	Used in prototype

Recommended Electrical Corrections Before Release

Improvement	Engineering purpose	Release action
0.1 uF bypass capacitor at AM26LS32 VCC/GND	Reduce supply-noise susceptibility and receiver glitches	Add adjacent to receiver supply pins and verify placement
Series base resistors on D10 and D13 transistor bases	Limit GPIO current and protect controller outputs	Select resistor value from interface-current requirement; document value
120 ohm termination across A+/A-, B+/B-, and Z+/Z-	Reduce differential-line ringing and reflections	Install where cable length/environment requires; verify signal quality
Controlled PCB/harness release	Replace temporary breadboard wiring and ambiguous splices	Issue schematic, BOM, assembly drawing, and continuity-test record

Mechanical Design Basis and Requirement Disposition

Requirement	Target	Source result	Disposition
Rotation precision	≤ 0.1 deg	No conclusive source result	Open / verify with calibrated method
Continuous rotation	≥ 360 deg	> 360 deg demonstrated	Pass
Translation precision	$\leq 1/32$ in	$1/32$ in observed	Pass
Load capacity	Original requirement 25 lb	40 lb static demonstration	Requirement exceeded in static demonstration; not a rated payload
Rotation rate	0 to ± 15 deg/s	Separate high-speed configuration referenced; low-speed characterization uses different configuration	Configuration-dependent / verify per release
Envelope	Approx. 12 x 12 in	12 x 12 in reported	Pass
Linear offset	± 3 in	± 3 13/32 in demonstrated	Pass

The 40 lb result is a static demonstration, not a certified dynamic payload rating. The original load requirement was 25 lb. Rated payload, fatigue life, shock, vibration, and environmental performance remain outside the demonstrated evidence.

Physical Wiring and Power-Up Sequence

Step	Condition / action	Required check
1	Power isolated	Verify driver supply is off, E-stop is accessible, and stage travel is clear.
2	Inspect switch configuration	Record microstep, current, and standby-current switch positions.

3	Connect encoder differential pairs	Maintain A+/A-, B+/B-, and Z+/Z- polarity; install required termination.
4	Connect controller interface	Verify seven-pin connector order, +5 V, and common ground before power.
5	Connect R725 STEP/DIR and motor	Confirm driver connector pinout, motor phases, and protective-earth arrangement.
6	Apply logic power	Confirm encoder counts and inactive STEP/DIR state.
7	Apply motor-driver power	Observe for unexpected motion or abnormal current; stop immediately if present.
8	Perform low-speed direction check	Command a small motion and verify physical and encoder directions agree.
9	Run acceptance test	Complete travel, E-stop, encoder, and commanded-motion checks.
10	Shutdown	Command zero, remove motor power, then logic power before service.

Hardware Safety and Acceptance

ID	Acceptance criterion	Verification
PL-AT-01	E-stop is accessible and removes or inhibits hazardous motion	Functional stop test
PL-AT-02	No exposed conductor, loose terminal, or reversed polarity is present before energization	Pre-power inspection
PL-AT-03	Encoder A/B direction agrees with commanded positive and negative travel	Low-speed direction test
PL-AT-04	Stage achieves at least +/-3 in travel without collision or binding	Measured travel test
PL-AT-05	Seven-pin connector continuity matches the controlled pinout	Continuity test
PL-AT-06	Microstep and current settings are recorded in the configuration record	Visual inspection and release record
PL-AT-07	Interface board ground plane is continuous and signal routing matches schematic intent	PCB review / continuity inspection
PL-AT-08	Static-load evidence is reported as demonstration only, not certified capacity	Document review